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THE INFLUENCE OF THE EARTH'S ROTATION UPON ROTATING BODIES AT ITS SURFACE.

BY W. FERREL.

In a late number of this Journal we endeavored to show that a moving body at the surface of the earth is subject to deflecting forces, depending upon the earth's rotation, which causes its motion relative to the earth to deviate a little from what it would be if the earth were at rest. It is proposed to show here, that the same deflecting forces operate upon the different parts of a rotating body at the earth's surface; so that if it be suspended so as to be free to turn about any axis passing through the center of gravity, the axis of revolution will not remain permanent, but change its direction in consequence of the action of these deflecting forces. This article is designed to be a supplement to the former, and consequently is based upon its fundamental equations.

If we put m, p and q for the sum of the moments of the forces which tend to turn the rotating body around the axes which are perpendicular respectively to the plane of the meridian, the prime vertical and the horizon, using the same notation as in the former article, and putting r', θ' and ϖ' for the values of r, θ and ϖ belonging to the center of gravity, we shall have

$$\begin{aligned} m &= \int r\,(r-r')\,\frac{dd\theta}{dt^2}\,dm - \int r\,(\theta-\theta')\,\frac{ddr}{dt^2}\,dm, \\ p &= \int r\,(r-r')\sin\theta\,\frac{dd\pi}{dt^2}\,dm - \int r\sin\theta\,(\varpi-\varpi')\,\frac{ddr}{dt^2}\,dm, \qquad [1] \\ q &= \int r^2\sin\theta\,(\varpi-\varpi')\,\frac{dd\theta}{dt^2}\,dm - \int r^2\sin\theta\,(\theta-\theta')\,\frac{dd\pi}{dt^2}\,dm. \end{aligned}$$

Let $\varrho =$ the distance of any particle dm of the rotating body from the axis of rotation;

$\alpha =$ The distance from the center of gravity of a plane passing through any particle dm, perpendicular to the axis of rotation;

$\psi =$ the angle of elevation of the axis above the horizon;

$\varphi =$ the angle between the plane of the meridian and a vertical plane passing through the axis, the angle being reckoned from the north toward the east;

$\mu =$ the angle between two planes passing through the axis, one of which is vertical, and the other passes through the particle dm.

We shall then have

$$\begin{aligned} r-r' &= -\alpha\sin\psi + \varrho\cos\psi\cos\mu, \qquad [2] \\ r\,(\theta-\theta') &= \alpha\cos\psi\cos\varphi + \varrho\sin\varphi\sin\mu + \varrho\sin\psi\cos\varphi\cos\mu, \\ r\sin\theta(\varpi-\varpi') &= -\alpha\cos\psi\sin\varphi + \varrho\cos\varphi\sin\mu - \varrho\sin\psi\sin\varphi\cos\mu. \end{aligned}$$

In obtaining the values of $\frac{dr}{dt}$, $\frac{d\theta}{dt}$ and $\frac{d\pi}{dt}$ from the preceding equations, $\frac{d\psi}{dt}$ and $\frac{d\phi}{dt}$ may be neglected in the present investigation, since they are exceedingly small in comparison with $\frac{d\mu}{dt}$; and $\frac{d\mu}{dt}$ may also be put equal v, the rotatory velocity of the rotating body. Hence we obtain

$$\begin{aligned} \frac{dr}{dt} &= -\varrho\cos\psi\sin\mu\,v, \\ r\frac{d\theta}{dt} &= \varrho\sin\varphi\cos\mu\,v - \varrho\sin\psi\cos\varphi\sin\mu\,v, \qquad [3] \\ r\sin\theta\frac{d\pi}{dt} &= \varrho\cos\varphi\cos\mu\,v + \varrho\sin\psi\sin\varphi\sin\mu\,v. \end{aligned}$$

By means of equations [2] and [3], together with equations [5] of the former article, neglecting all terms which are 0 by integration, equations [1] may be reduced to the following forms:

$$\begin{aligned} m &= -2\,n\,v\,(\sin\theta\sin\psi - \cos\theta\cos\psi)\,A, \\ p &= -2\,n\,v\cos\theta\cos\psi\sin\varphi\,A, \qquad [4] \\ q &= -2\,n\,v\sin\theta\cos\psi\sin\varphi\,A, \end{aligned}$$

in which $A = \int\varrho^2\sin^2\mu\,dm$.

If in these equations we put $\varphi = 0$ and $\psi =$ the complement of θ, that is, if we suppose the axis of the rotating body to be parallel with the axis of the earth, we shall have $m = 0$, $p = 0$ and $q = 0$; and hence the axis of the rotating body is in this case permanent. In all other positions the deflecting forces will cause it to change its direction, and its motion may be determined by the preceding equations. If the axis of the rotating body is free to move in the plane of the meridian only, its motion may be determined by the first of equations [4], and is oscillatory about the position in which it is parallel to the axis of the earth. If it is free to move in the prime vertical only, its motion is determined by the second of equations [4], which is the equation of the pendulum, and the oscillations are about the perpendicular position of the axis. Lastly, if it is free to move horizontally only, its motion is determined by the last of equations [4], which is likewise the equation of the pendulum, and the oscillations are about its position when in the plane of the meridian. These deflecting forces, however, are so small, and, in any experiments with the most delicate apparatus, the friction in comparison is so great, that these oscillations are not observed, but only a tendency of the axis of rotation to assume the position in which all the forces are in equilibrium, which, in the general case in which the axis of rotation is free to turn in any direction, is a position parallel with the axis of the earth.

These deductions from theory are completely verified by some very delicate experiments which have been made by Foucault with a peculiar form of the gyroscope, an account of which has been given by M. J. Nicklès in the *American Journal of Science and Arts*, Vol. xv, p. 263. See also Vol. xix, p. 141. It is, however, stated by M. Nicklès, that the angular movement about the vertical axis may be shown to be proportional to the sine of the latitude; whereas, from the last of equations [4], it must be as the sine of the polar distance, and this is also evident from other considerations. It is well known that a descending body, on account of the earth's rotation, is deflected towards the east; and, on the contrary, an ascending body, towards the west. Hence when the torus of Foucault's apparatus is placed in the plane of the meridian, the side which descends is deflected towards the east, and the side which ascends, towards the west; and hence there is a torsion of the torus when free to move about a vertical axis, and the force which produces it is well known to be as the sine of the polar distance. The last of equations [4] is an expression of this force, and there is no other which tends to turn the torus or rotating body around a vertical axis.

Nashville (Tennessee), Jan. 12, 1858.

ORBIT OF VIRGINIA.

BY JOHN N. STOCKWELL.

The elements of *Virginia* given by Mr. Ferguson (*Astr. Journal*, N° 108), compared with the observations, give the following residuals:

Date.	C.—O. $\Delta\alpha$	$\Delta\delta$	
1857 Oct. 7.0	$+2''.75$	$-2''.58$	8 observations.
Nov. 10.5	$+3'\ 0.72$	$+1'\ 50.90$	4 "
Dec. 12.5	$+13\ 42.94$	$+7\ 29.64$	4 "

Hence we get the following normal places:

1857	⑤⓪ α	⑤⓪ δ	
Oct. 7.0	13° 56′ 18″.55	+3° 40′ 56″.98	Mean Equinox 1857 Jan. 1.0
Nov. 10.5	10 30 8.58	+1 15 55.60	
Dec. 12.5	14 41 27.12	+2 57 49.78	

These normals give the following

ELEMENTS OF *VIRGINIA*.

1857 Oct. 7.0 Washington M. T.

$$
\begin{aligned}
M &= 2^\circ\ 14'\ 34''.27 \\
\Omega &= 173\ 30\ 49.32 \\
\pi &= 10\ \ 6\ 56.87 \\
i &= 2\ 47\ 50.70 \\
\varphi &= 16\ 39\ 55.33 \\
\log a &= 0.4229928 \\
\log \mu &= 2.9155174
\end{aligned}
$$

(Ω, π: Mean Eq. 1857 Jan. 1.0)

These elements represent the middle observation as follows:

C.— O.

$$\Delta\alpha = -0''.08; \quad \Delta\delta = -0''.03.$$

It is my intention to compute the perturbations, and thus furnish more accurate elements, and an ephemeris for the next opposition.

ELEMENTS OF THE FIRST COMET OF 1858.

BY T. H. SAFFORD.

From observations at Cambridge, Jan. 4 and Feb. 6; and at Altona, Jan. 21.

$$
\begin{aligned}
\omega &= 206^\circ\ 48'\ 58''.8 \\
\varphi &= 55\ \ 6\ 31.2 \\
\Omega &= 269\ \ 3\ 25.5 \quad \text{M. Equinox 1858.0} \\
i &= 54\ 24\ 17.2 \\
\log a &= 0.756228
\end{aligned}
$$

The period is 13.625 years, or nearly $\frac{1}{5}$ of the interval ($68^y\ 26^d$) since the last observed perihelion passage.

All the observations are satisfied within 1″.